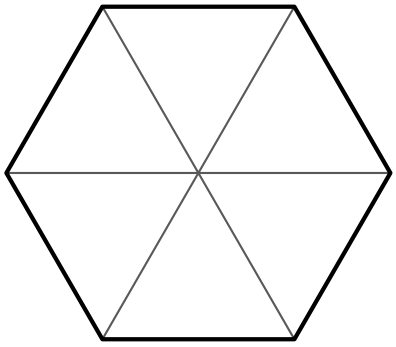
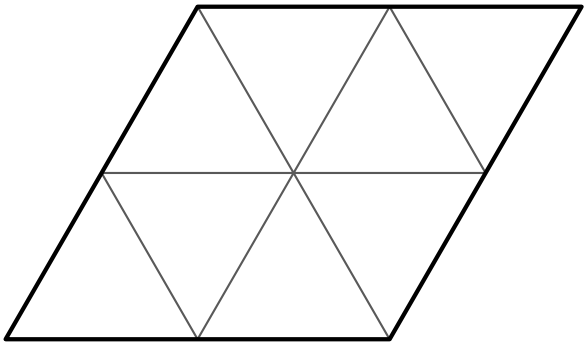
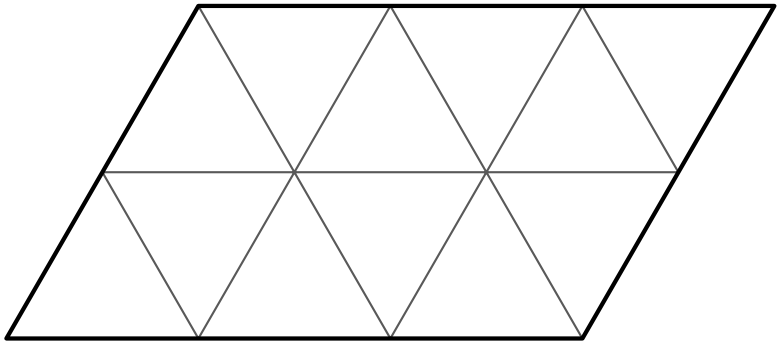
Problem 1: Play this game with a partner. Take turns placing one blue rhombus on the board so that it covers two small triangles and does not overlap another rhombus. The player who cannot place a rhombus loses. Play several games on each board below, and circle the player who can always win there.



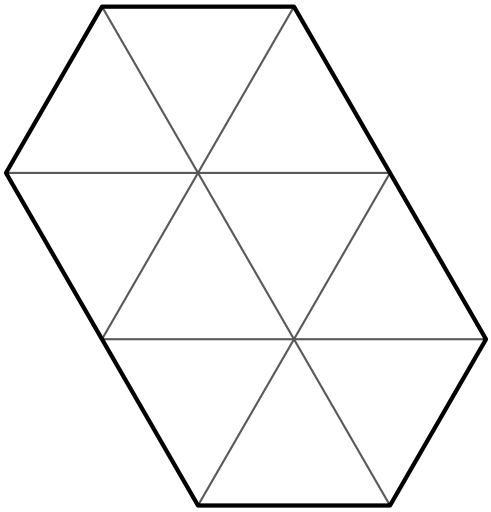
1st 2nd



1st 2nd

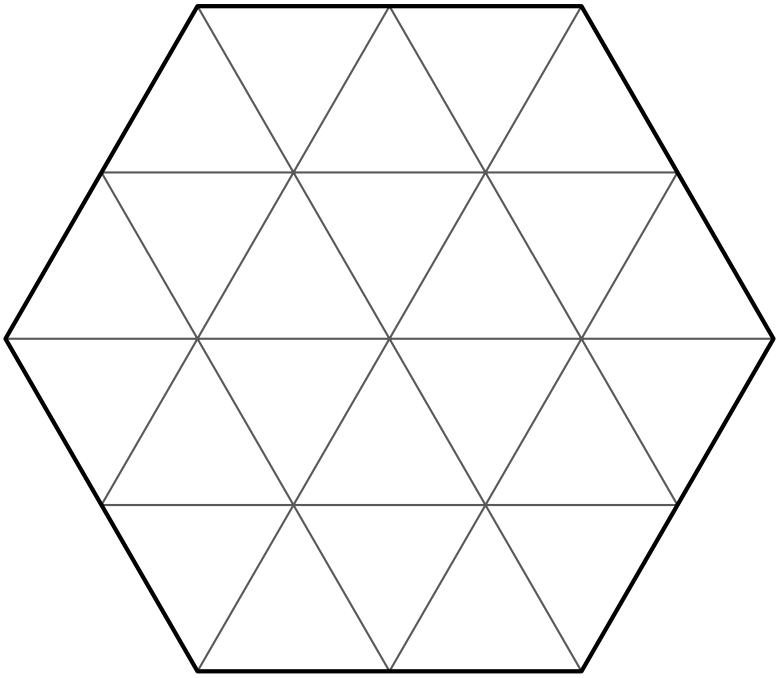


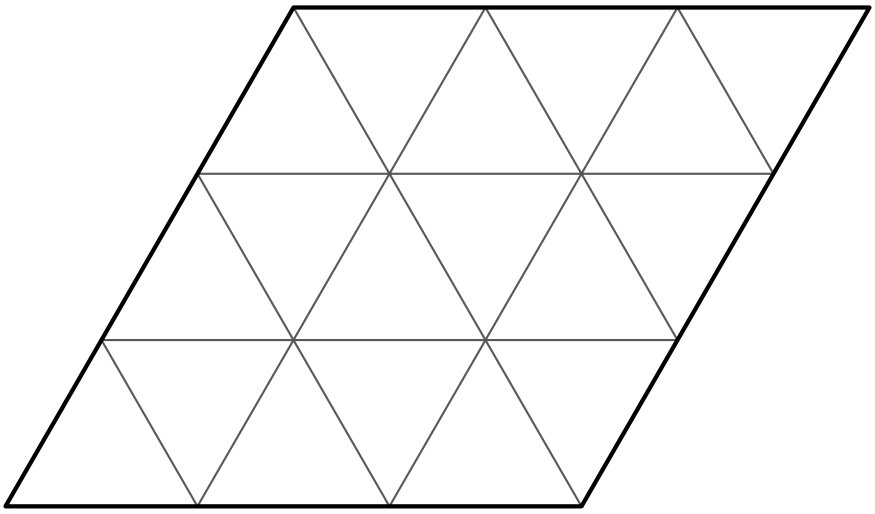
1st 2nd



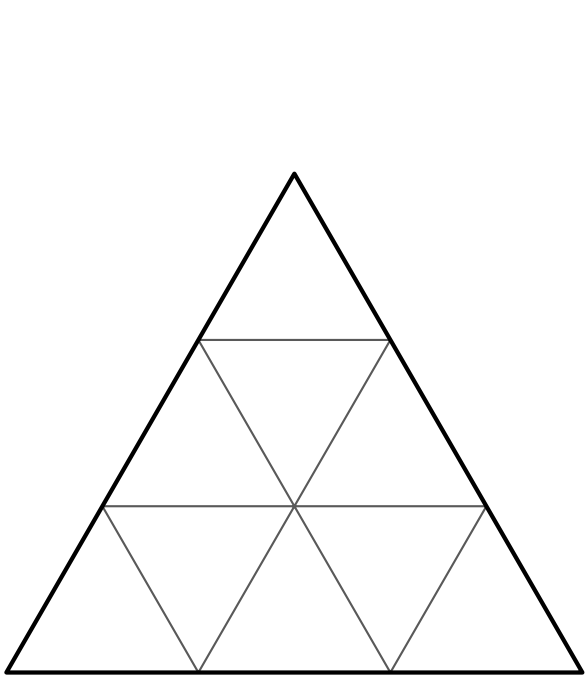
1st 2nd

Problem 2: Play the game on these two boards. On each board, find a way for one of the players to win every game, and explain why it always works.

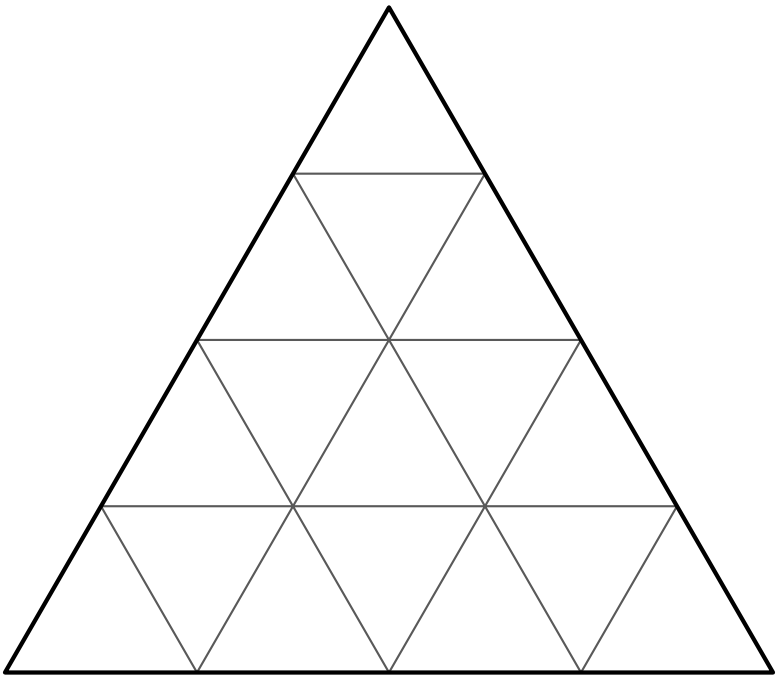




Problem 3: Play the game from Problem 1 on these two triangles. Circle the player who can always win on each one.

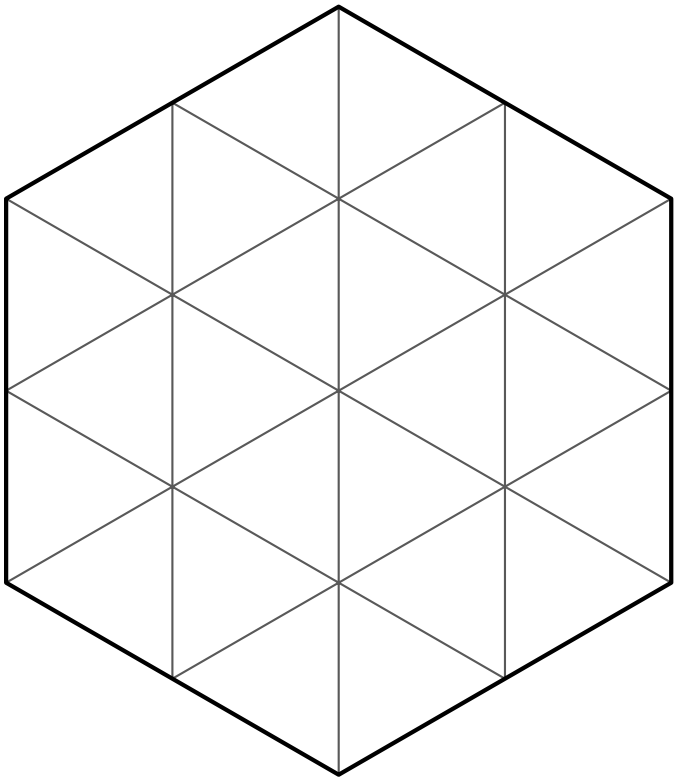


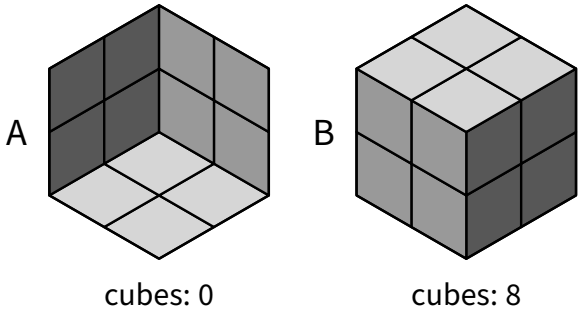
1st 2nd



1st 2nd

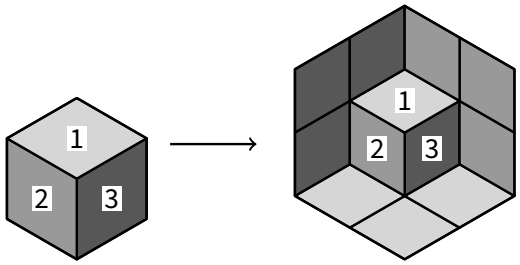
Problem 4: Fill the hexagon below with small blue rhombi. Shade a copy of your filling the way A and B are shaded, then build the pile of snap cubes it shows in the corner of the box and write how many cubes it has. Do this for as many different fillings as you can.



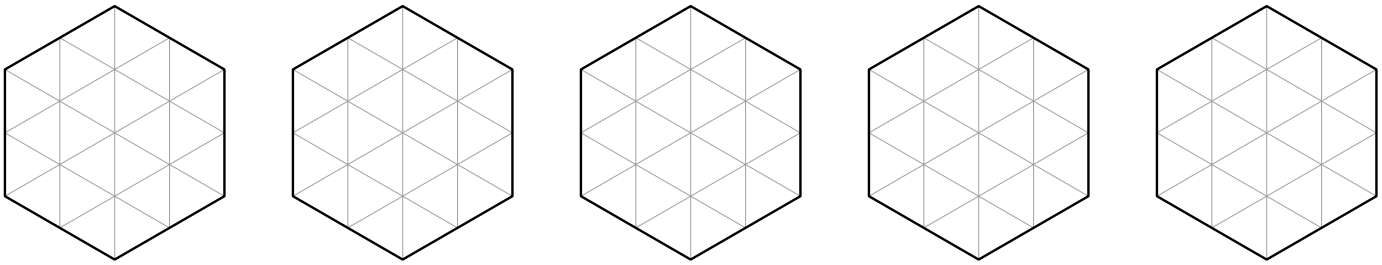


cubes: 0 cubes: 8

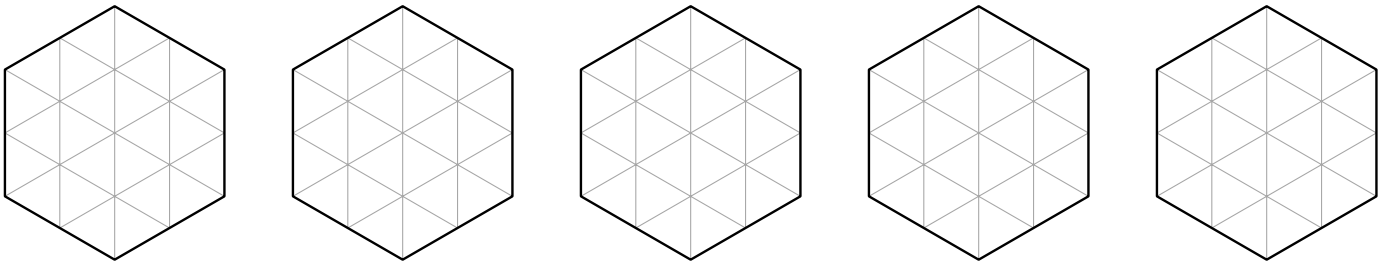
Add one cube to A.



one cube cubes: 1



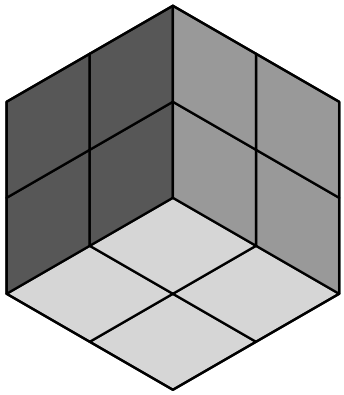
cubes: _____ cubes: _____ cubes: _____ cubes: _____ cubes: _____



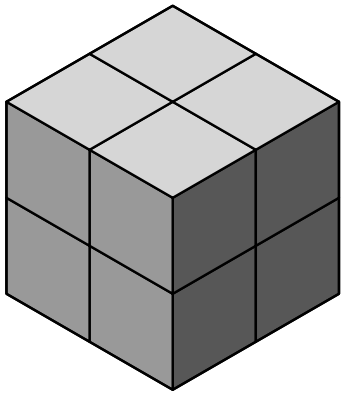
cubes: _____ cubes: _____ cubes: _____ cubes: _____ cubes: _____

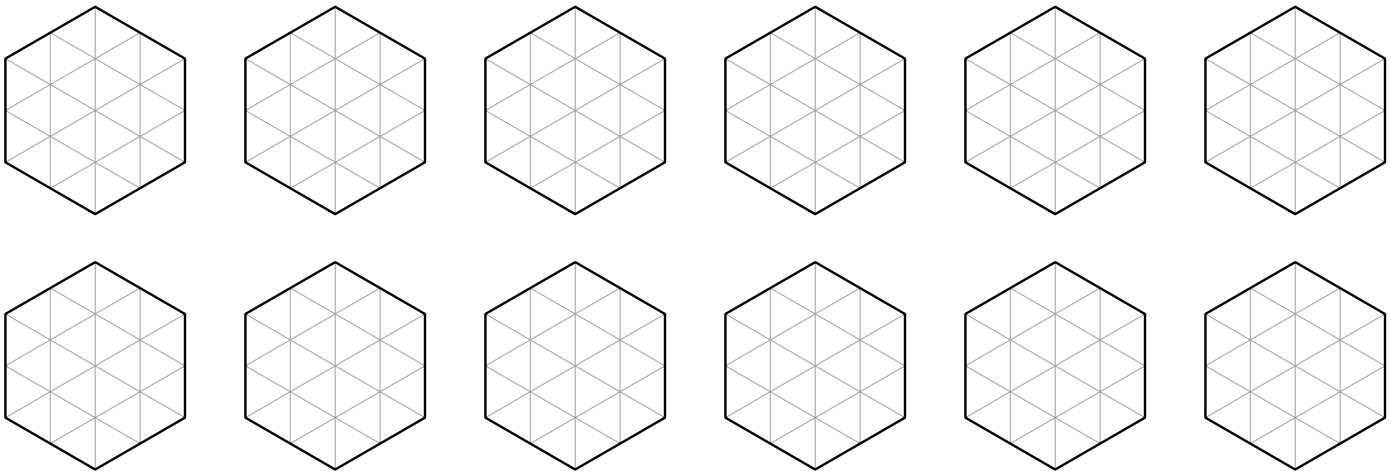
Problem 5: A flip takes three rhombi that make a small hexagon and puts them back the other way. Change filling A into filling B by flips, using as few flips as you can, and explain why fewer flips cannot work. Can a route of flips start at a filling and come back to it after an odd number of flips? Explain.

A

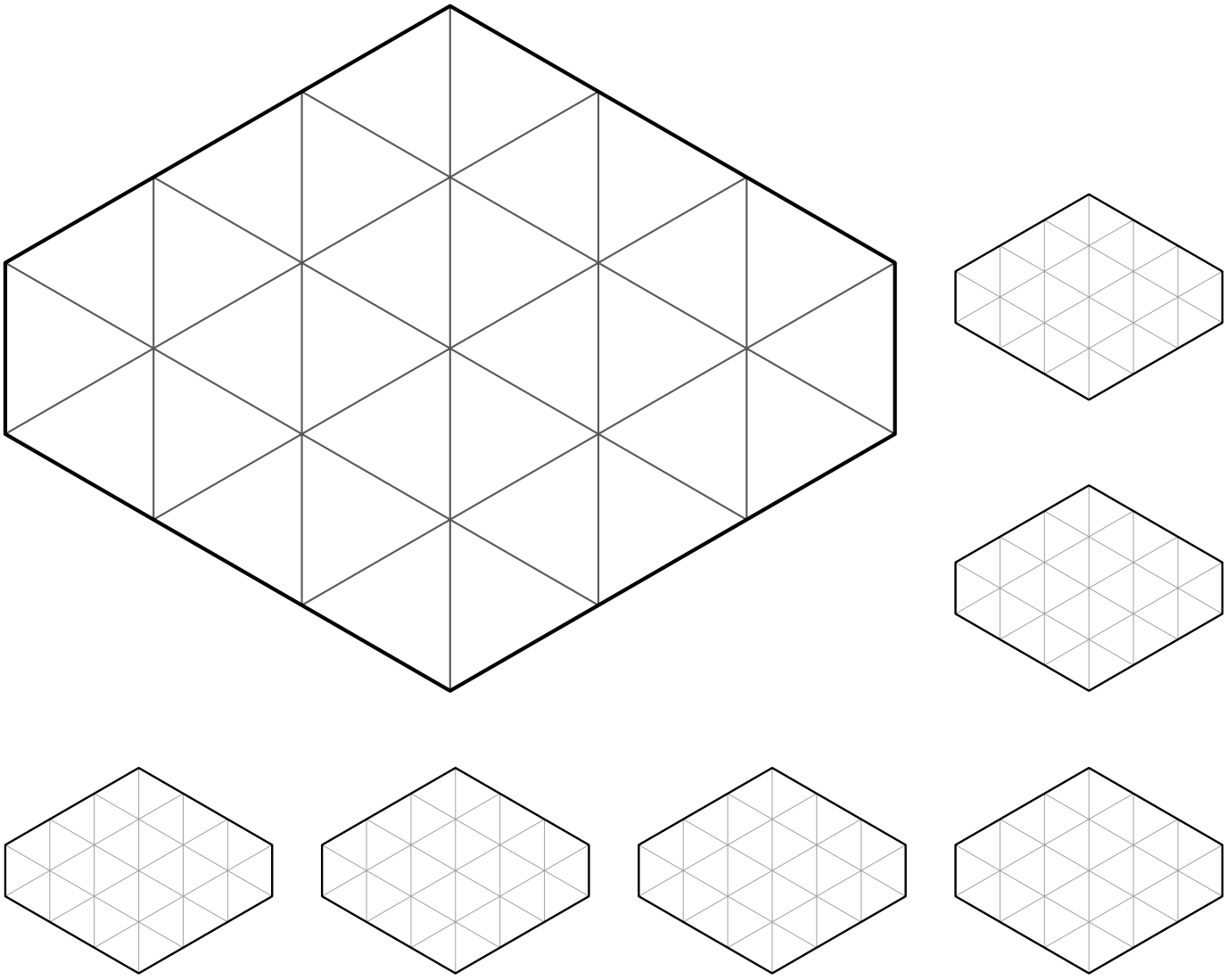


B

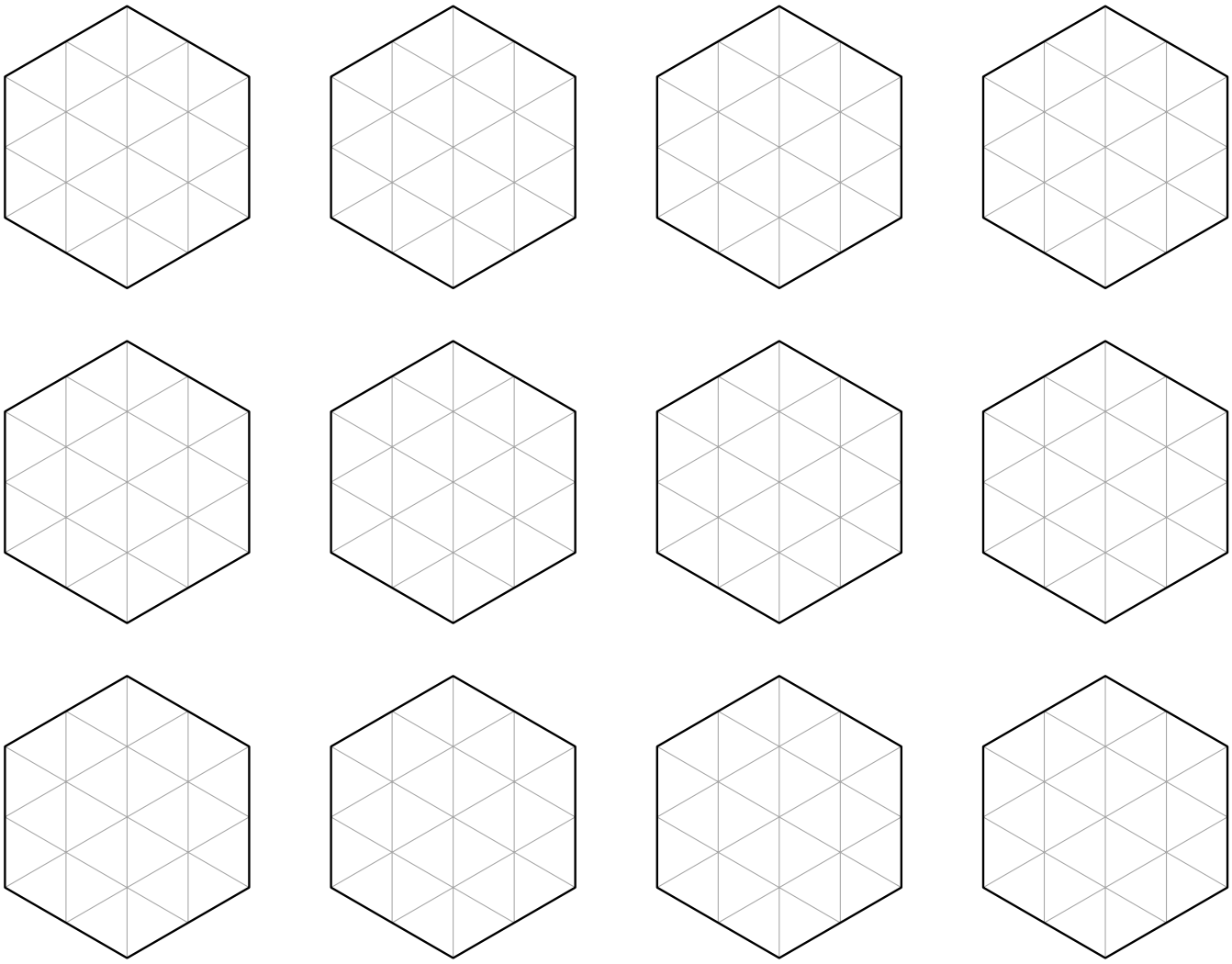




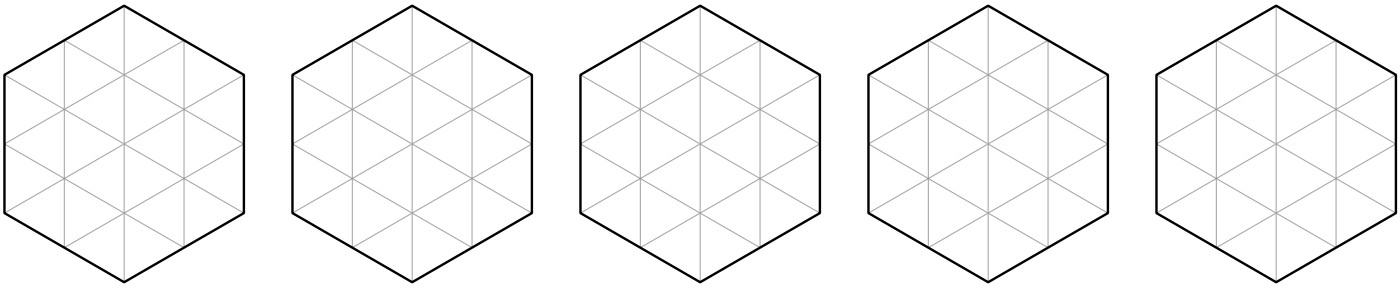
Problem 6: Fill this hexagon with small blue rhombi in several ways, and shade each filling on a small copy as in Problem 4. How many rhombi of each shade does each filling have? What is the smallest number of flips that changes the filling with no cubes into the filling with the most cubes? Explain both answers.



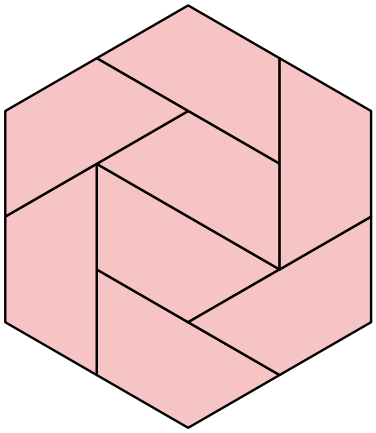
Problem 7: Fill the hexagon from Problem 4 with small red trapezoids. Find every way to do it, draw each one, and explain how you know you have found them all.



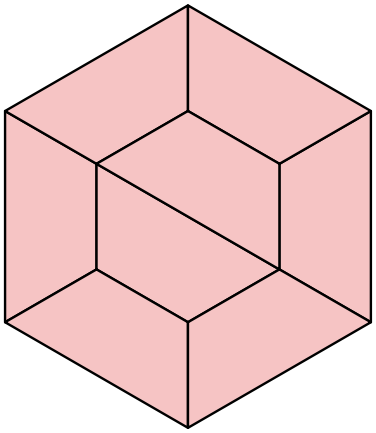
Problem 8: A move picks up some of the trapezoids in a filling from Problem 7 and puts them back in a different way. Find a route of moves, each picking up two trapezoids, that starts at a filling and comes back to it after an odd number of moves. Can filling C be changed into filling D by moves that each pick up five trapezoids or fewer? Explain.



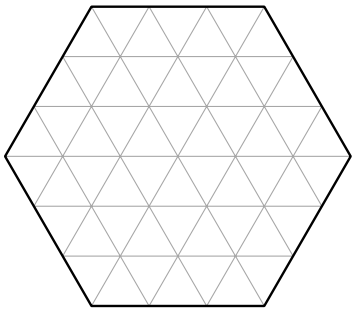
C



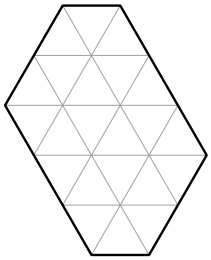
D



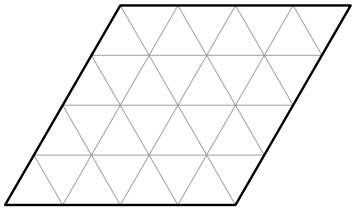
Problem 9: Decide who can always win the game from Problem 1 on each of these boards, and explain how that player wins.



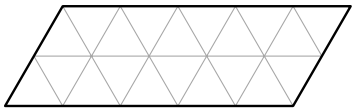
1st 2nd



1st 2nd

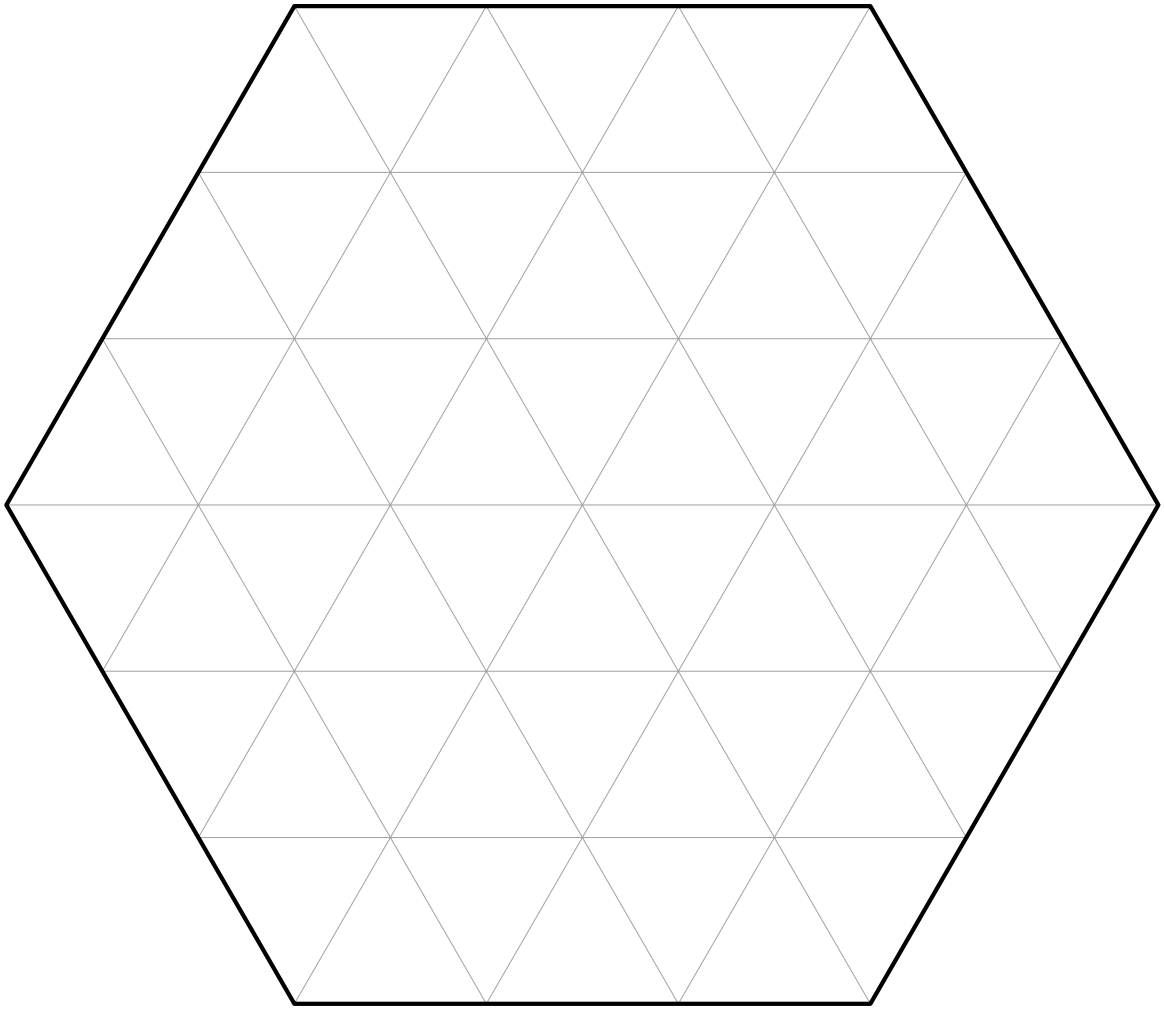


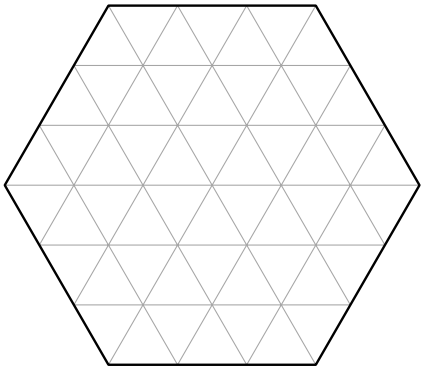
1st 2nd



1st 2nd

Problem 10: Fill this big hexagon with as few pieces as you can, using only the small green triangles, blue rhombi, red trapezoids and yellow hexagons. Draw your best filling on the small copy and write how many pieces it has.





pieces